HIGH PRESSURE PROCESS WATER PUMPS
2000-F17-PID-PR-10002
HP PROCESS WATER
17-0174-PW-PE16-160

MILL WATER PUMPS
2000-F24-PID-PR-10001
PROCESS WATER
17-0360-PW-PE8-225
17-0361-PW-SS1-200
200VBFD1S2 F17-10837
200VBFD1S1 F17-10838
50VB03S1 F17-10841
150VBFD1S1 FC
200VBFD1S1 F17-10839
200VCD1S1 F17-10842
TO GRADE
17-0925-PW-SS3-50
200VBFD1S1 F17-10840

FLOCCULANT DOSING PUMPS
2000-F72-PID-PR-10004
FLOCCULANT
17-0371-FL-PE8-90
17-0372-FL-MS1-80
80VBD1C1 F17-10843
80VBD1C1 F17-10844
25VB03S1 F17-10845
17-0275-FL-MS1-25
TO GRADE
F17-XM-413
FLOTATION TAILINGS THICKENER FLOCCULANT INLINE MIXER
17-0130-PW-SS3-50
17-0267-PW-PE16-63
300x200

FLOTATION AREA SUMP PUMP 2
2000-F16-PID-PR-10003
SPILLAGE
16-1208-FD-PE10-160

FLOTATION AREA SUMP PUMP 3
2000-F16-PID-PR-10004
SPILLAGE
16-1214-FD-PE10-160

CLEANER AREA SUMP PUMP
2000-F18-PID-PR-10001
SPILLAGE
18-1216-FD-PE10-110

FLOCCULANT AREA SUMP PUMP
2000-F72-PID-PR-10004
SPILLAGE
72-1076-FL-PE8-90

FLOTATION TAILINGS THICKENER UNDERFLOW PUMP
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
17-0383-CON-PE10-500

MILL WATER PUMPS
2000-F24-PID-PR-10001
PROCESS WATER
17-0187-PW-PE8-500
500VBFD1L2 F17-10846

FLOTATION CELLS DRAIN PUMP
2000-F16-PID-PR-10003
SCAVENGER TAILS
16-1290-FS-PE10-280

FLOTATION AREA SUMP PUMP 1
2000-F16-PID-PR-10001
SPILLAGE
16-1206-FD-PE10-160

RAW WATER PUMPS
2000-F24-PID-PR-10002
RAW WATER
17-0300-BW-PE8-63
17-0975-BW-SS3-50
50VC01S1 F17-10847
17-1424-BW-SS3-25
25VB03S1 F17-14149
25VB03S1 F17-10848
25VB03S1 F17-10849
25VB03S1 F17-14109
F17-XM-572 FLOTATION TAILINGS THICKENER MANTA SUB
DETAIL 5

FLOTATION DISCHARGE BOX
2000-F16-PID-PR-10003
SCAVENGER TAILS
16-1248-FS-RL1-1200
DN1200
F17-SA-416 FLOTATION TAILINGS THICKENER FEED SAMPLER
VENDOR PACKAGE 2000-ME074-J02-10007
17-1742-FS-RL1-1200
17-1743-FS-RL1-80
80VKGD1L1 F17-10851
1200VKGD1L2 FL
F17-BX-410 FLOTATION TAILINGS THICKENER RECIRCULATION FEED BOX
17-0497-FS-RL1-1200
F17-HU-422 FLOTATION TAILINGS THICKENER FEED BOX VALVE HYDRAULIC POWER UNIT
TI 17141
PI 17140
LI 17142
F17-XM-474 F17-XM-471 F17-XM-613 F17-XM-472 F17-XM-614 F17-XM-473
FLOTATION TAILINGS THICKENER OVERFLOW LAUNDER DART VALVES (6 OFF)
(6 OFF)
F17-PP-459 FLOTATION TAILINGS P-DUC PUMP
F17-TH-401 FLOTATION TAILINGS THICKENER
THICKENER FEED STANDPIPE NOTE 6

17-0976-PW-SS2-100
17-0984-PW-SS2-100
17-0985-PW-SS2-100
17-0986-PW-SS2-100
17-0987-PW-SS2-100
17-0988-PW-SS2-100
17-0982-PW-PE16-160
17-0983-PW-PE16-160
100VBFE1S1 F17-10824
100VBFE1S1 F17-10826
100VBFE1S1 F17-10828
100VBFE1S1 F17-10830
100VBFE1S1 F17-10832
100VBFE1S1 F17-10834
17-1487-PW-SS3-50
17-1486-PW-SS3-50
17-1485-PW-SS3-50
17-1484-PW-SS3-50
17-1483-PW-SS3-50
17-1482-PW-SS3-50
50VB03S1 F17-10825
50VB03S1 F17-10827
50VB03S1 F17-10829
50VB03S1 F17-10831
50VB03S1 F17-10833
50VB03S1 F17-10835
F24-XM-451
F24-XM-452
F24-XM-453
F24-XM-454
F24-XM-455
F24-XM-456
FLOTATION TAILINGS THICKENER WATER MONITORS (6 OFF) NOTE 7
MC-002

PI 17002
PT 17002
50VBE1S1 F17-10823
FI 17003
FIC 17001
FIT 17001
FE 17001
FCV 17001
FIT 17003
FE 17003
DILUTION FACTOR
DETAIL 1

MANTA SUB START/STOP
HS 17038
MANTA SUB RUNNING
XI 17038A
MANTA SUB FAULT
XA 17038B
MANTA SENSOR FAILED TO RETURN HOME ALARM
XA 17038
NOTE 4
XA 17060 EMERGENCY STOP
ZAH 17014 MAX. RAKE LEVEL
ZAL 17015 MIN. RAKE LEVEL

ZS 17010 TURBIDITY
AI 17019
LIT 17012
LE 17012
AIT 17019
AE 17019
BED LEVEL METER MANTA SUB
RAKE HEIGHT LO
RAKE HEIGHT HI
RAKE UNDERSPEED SWITCH
GEARBOX LEVEL SWITCH LOW
GEARBOX LEVEL SWITCH LOW
RAKE HEIGHT POSITION
ZSL 17015
ZSH 17014
SSL 17016
LSL 17018
LSL 17017
ZIT 17013
ZE 17013
F17-XM-572 MANTA SUB
F17-HU-406 FLOTATION TAILINGS THICKENER RAKE DRIVE HPU
F17-HU-414 FLOTATION TAILINGS THICKENER RAKE LIFT HPU
VENDOR PACKAGE 2000-ME0015-E02-50001 TO 50006

XI 17066 24VDC POWER HEALTHY
SV 17033 RAKE DRIVE FORWARD SOLENOID RELAY
SV 17034 RAKE DRIVE REVERSE SOLENOID RELAY
SV 17048 RAKE LIFT RAISE SOLENOID RELAY
SV 17047 RAKE LIFT LOWER SOLENOID RELAY
XC 17076 RAKE DRIVE RUNNING FEEDBACK
XC 17057 LOCAL MODE ENABLE FEEDBACK
NI 17030 TORQUE INDICATION
XC 17079 85% TORQUE FROM DCS TO LCP
XC 17083 50% TORQUE FROM DCS TO LCP
XC 17080 40% TORQUE FROM DCS TO LCP
XC 17084 30% TORQUE FROM DCS TO LCP

HS 17065A RAKE LIFT LOCAL RAISE REQUEST
HS 17065B RAKE LIFT LOCAL LOWER REQUEST
HS 17063 REVERSE JOG LOCAL START REQUEST
XC 17082 ALARM ACKNOWLEDGE FROM DCS TO LCP
XC 17062 REVERSE JOG ACKNOWLEDGE FROM DCS TO LCP
HS 17061 RAKE DRIVE FORWARD LOCAL START REQUEST
HS 17064 RAKE DRIVE LOCAL STOP REQUEST
PDAH 17036 RAKE DRIVE OIL TANK COOLING FAN OUTLET FILTER DIFF PRESSURE (WORKING)
PDAH 17028 RAKE DRIVE OIL TANK PUMP OUTLET FILTER DIFF PRESSURE
PDAH 17037 RAKE DRIVE OIL TANK COOLING FAN OUTLET FILTER DIFF PRESSURE (STANDBY)
PDAH 17031 RAKE DRIVE OIL TANK OUTLET OIL DIFF PRESSURE
ZI 17013 RAKE LIFT POSITION
TAH 17022 RAKE DRIVE OIL TANK TEMPERATURE HIGH

TAHH 17024 RAKE DRIVE OIL TANK TEMPERATURE HIGH
LALL 17023 RAKE DRIVE OIL TANK LEVEL LOW LOW
LAL 17021 RAKE DRIVE OIL TANK LEVEL LOW
LAL 17041 RAKE LIFT OIL TANK LEVEL LOW
TAH 17042 RAKE LIFT OIL TANK TEMPERATURE HIGH
LALL 17043 RAKE LIFT OIL TANK LEVEL LOW LOW
TAHH 17044 RAKE LIFT OIL TANK TEMPERATURE HIGH HIGH
PDAH 17051 RAKE LIFT OIL TANK PUMP OUTLET FILTER DIFF PRESSURE
LAL 17017 GEAR REDUCER OIL LEVEL LOW
LAL 17018 GEAR REDUCER OIL LEVEL LOW
SAL 17016 RAKE ZERO SPEED
ZAO 17026 F17-PO-408 THICKENER RAKE DRIVE PUMP SUCTION VALVE POSITION SWITCH OPENED

F17-VL-458 FLOTATION TAILINGS THICKENER OVERFLOW LAUNDER
F17-BX-464 FLOTATION TAILINGS THICKENER OVERFLOW BOX
F17-XM-572 FLOTATION TAILINGS THICKENER MANTA SUB
F17-VL-490 FLOTATION TAILINGS THICKENER OVERFLOW PIPE LAUNDER 1
F17-VL-491 FLOTATION TAILINGS THICKENER OVERFLOW PIPE LAUNDER 2
MECH
PROCESS WATER TANK 2000-F24-PID-PR-10001 FLOTATION TAILINGS THICKENER OVERFLOW
FIC17121 2000-F17-PID-PR-10002 SIGNAL
BED MASS METER
PE 17011
BED PRESSURE PIC 17011
50VB03S1 F17-10857
50VB03S1 F17-10858
50VB03S1 F17-10859
50VB03S1 F17-10860
50VB03S1 F17-10861
50VB03S1 F17-10862
17-0498-CON-RL1-300
300VKGD1L1 F17-10863
TO GRADE
DN50
1:100
17-0364-CON-RL1-600
17-0374-CON-RL1-600
17-0365-CON-RL1-600
17-0377-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 1 2000-F17-PID-PR-10002 FLOTATION TAILINGS THICKENER UNDERFLOW
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2 2000-F17-PID-PR-10002 FLOTATION TAILINGS THICKENER UNDERFLOW
FLOTATION PARTICLE SIZE ANALYSER 2000-F16-PID-PR-10005 FINAL TAILS SAMPLE
16-0661-FS-PE10-63

CCTV F17-CAM-001
F17-TAH-17039
SP SC-034
NOTE 5
VSD
DN150
THICKENER FEED STANDPIPE NOTE 6
F17-PP-459 FLOTATION TAILINGS P-DUC PUMP
F17-TH-401 FLOTATION TAILINGS THICKENER
SP HS-100
SP MC-001
SP MC-002
SP @ BRIDGE OF THICKENER F17-TH-401
17-0373-FL-SS1-300
25VB03S1 F17-10821
50VB03S1 F17-10822

SP HS-056
SP HS-025
SP SC-034
XV 17009
HS 17009
ID25mm
DN1200
1.5°
SP HS-105
DN50
PE8 SS1
PE8 MS1
PE8 SS3
PE10 RL1
PE16 SS2 SS3
PE16 SS3
RL1 PE10
90x63

F16-HP-456 FLOTATION TAILS SAMPLE HOPPER
16-3072-FS-PE10-110
TO GRADE
16-0496-FS-PE10-63
LIC 17004
LT 17004
LI 17004
LAH 17081
LSH 17081
NOTE 7
F17-PS-427 FLOTATION TAILINGS THICKENER AREA SUMP PUMP 1
17-0367-FD-RL1-200
17-0368-FD-PE10-225
50VB03S1 F17-10853
200VKGD1L1 F17-10854
50VB03S1 F17-10852
200x100
DETAIL 3
HS 17005
XV 17005
80VKGD1L1 FL
16-0660-FS-RL1-80
16-2140-FS-RL1-80
80x65
DETAIL 4
HS 17006
XV 17006
HS 17007
XV 17007
50VKGD1L1 FO
16-1515-FS-SS3-50
16-1448-FS-SS3-50
ZAH 17144
PAH 17143
BURST
XA 17008
XS 17008
SP HS-074
F16-PP-634 FINAL TAILS SAMPLE PUMP

NOTES:
1. REFER TO CONTROL PHILOSOPHY FOR COMPREHENSIVE LIST OF PROCESS INTERLOCKS AND CONTROL FUNCTIONS.
2. STAGE 2.
3. TIE IN POINT LOCATED ON F17-BX-410.
4. SIGNAL FROM FLOTATION THICKENER LCP TO MCC/DCS.
5. FLOCCULANT IS BEING DISTRIBUTED WITHIN VENDOR PACKAGE.
6. THERE ARE 3 OFF SPARE NOZZLES (DN100, DN200 & DN60 AVAILABLE).
7. FLANGED ASME B16.5 CLASS 150.

DRG No.	REFERENCES
2000-F00-PID-PR-10007	LEGEND SHEET 7 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10006	LEGEND SHEET 6 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10005	LEGEND SHEET 5 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10004	LEGEND SHEET 4 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10003	LEGEND SHEET 3 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10002	LEGEND SHEET 2 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10001	LEGEND SHEET 1 - PIPING AND INSTRUMENTATION DIAGRAM

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2	12SEP25	LAUNDERS AND GENERAL REVISIONS ADDED	AH	MM	JBe	DA	DM
1	19NOV24	RE-ISSUED FOR CONSTRUCTION	NB	JB	JB	LS	DM
0	06SEP24	APPROVED FOR CONSTRUCTION	AK	JB	JB	LS	DM

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PRIMERO PROJECT NUMBER: 23517

PROJECT: NORTHERN STAR RESOURCES LTD KCGM GROWTH PROJECT
TITLE: FLOTATION TAILINGS PRE-LEACH THICKENING FLOTATION TAILINGS THICKENING - SHEET 1 - STAGE 1 PIPING AND INSTRUMENTATION DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NTS	2000-F17-PID-PR-10001	2

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